Worked solutions — class questions and practice

Only consult these solutions after completing the relevant classwork. Check and correct your own reasoning. Homework tasks and homework answers are not included.

Part A — Thinking like a scientist

1. Physics helps us measure and describe motion, predict what happens next, explain changes and design safer vehicles.

2. The quantity and unit are missing; if the number represents velocity, a direction is also needed.

3. A quantity is a measurable property; its symbol is shorthand; its SI unit states how it is measured.

Part B — Eight quantities

4. Distance is the length of the route actually travelled. Displacement is the directed change from the start to the finish.

5. Speed describes how fast. Velocity also specifies direction.

6. Average speed covers the whole interval: total distance ÷ elapsed time. A speedometer gives speed at an instant.

7. Yes. Turning changes the direction of velocity, so the car has acceleration.

Part C — Scalars and vectors

Table. Scalar: distance, speed, average speed. Vector: displacement, velocity, average velocity, acceleration.

8. Speed is scalar and cannot be negative. With right chosen positive, leftward velocity is negative.

9. The position is −15 m relative to the specified origin (right positive).

Part D — Instant and average

10. (a) is instantaneous; (b) is an average over an interval.

11. Distance is one lap, displacement 0; average speed is positive and average velocity is zero.

Part E — Six journeys

12. C accelerates from rest; D brakes to rest; both cover equal distance and elapsed time.

13. Velocity can be negative to show negative direction, but speed is a scalar magnitude.

14. Both have zero displacement and zero average velocity for the full chosen interval.

15. Velocity; speed; direction.